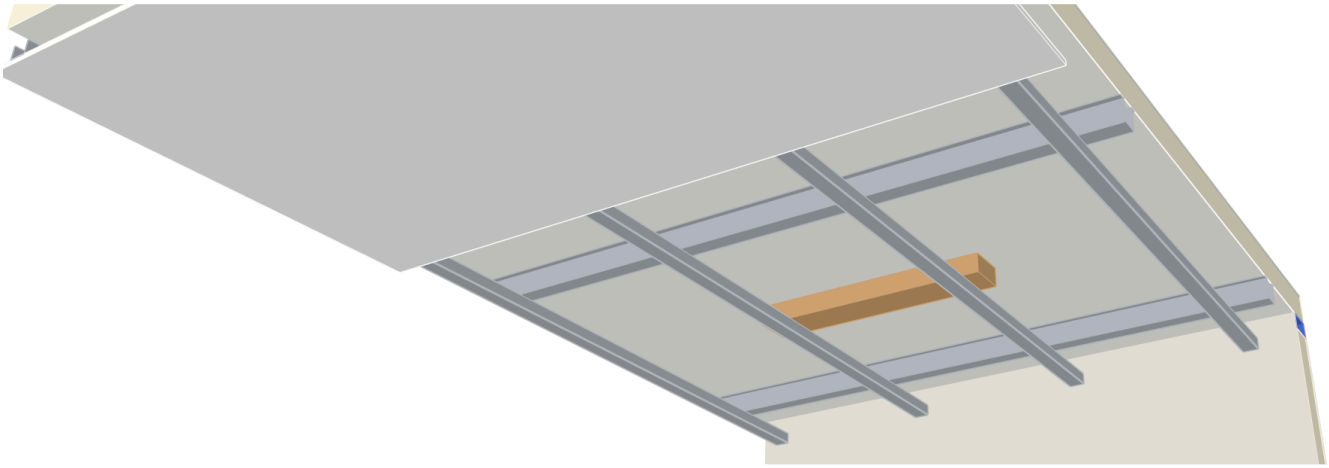
Suspended ceilings under Insulspan roof panels

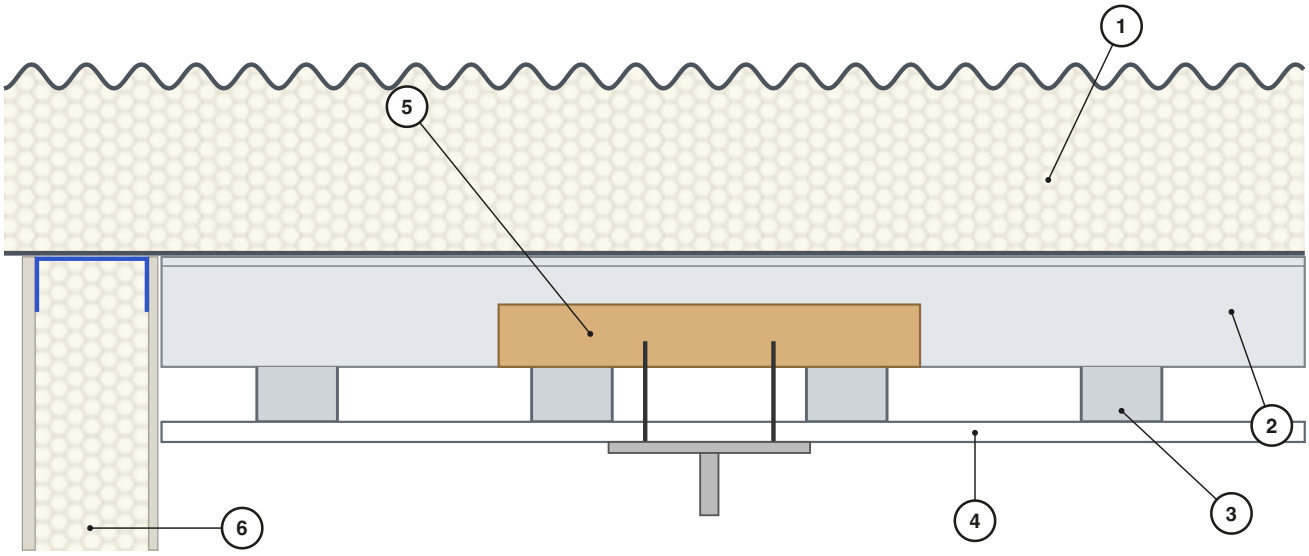
A plaster ceiling on top hats and battens under the roof panel, with a fixing block for ceiling fans.

A plaster ceiling suspended under the roof panel, on top hats and ceiling battens, gives a lined ceiling with a space above it, and a solid fixing block for a ceiling fan.



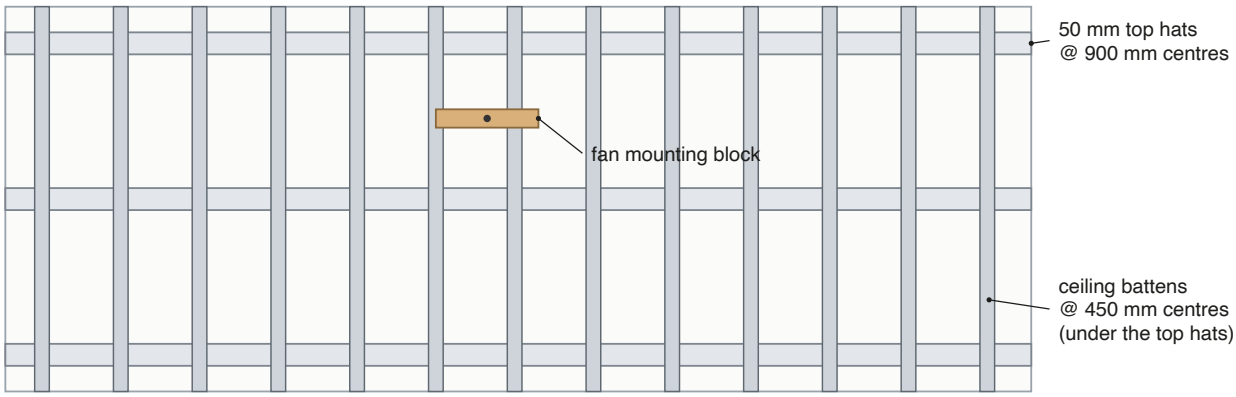
Top hats, battens and the fan mounting block under the roof panel (lining cut back).

1 Section



1. Roof panel: 0.42 mm corrugated sheet, EPS foam, 0.5 mm lining sheet
2. 50 mm top hats at 900 mm centres, fixed to the underside of the roof panel
3. Standard ceiling battens at 450 mm centres, fixed to the top hats
4. Plaster lining, fixed to the battens
5. Fan mounting block (566 mm), the fan fixed into it with 50 mm screws
6. External wall with steel top channel

2 Plan



3 Install steps

1. **Fix the 50 mm top hats** to the underside of the roof panel at 900 mm centres.
2. **Fix the ceiling battens** to the top hats at 450 mm centres.
3. **Fit a fan mounting block** (566 mm long) across the battens wherever a ceiling fan goes.
4. **Fix the plaster lining** to the battens, then fix the fan into the block with 50 mm screws.

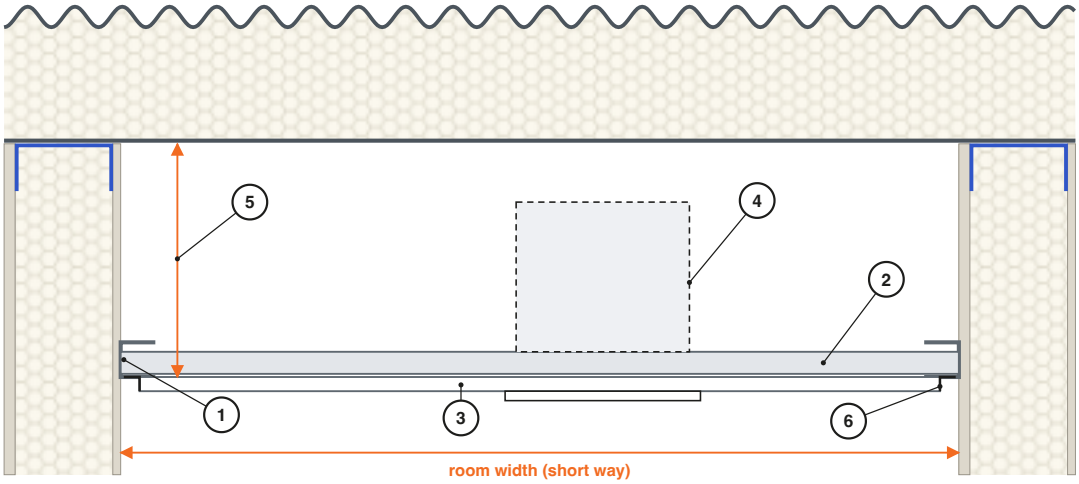
4 Bathroom heat / fan / light units (IXL Tastic)

- A Tastic needs **210–245 mm** of space above the ceiling, depending on the model. A 50 mm top hat and batten ceiling is not deep enough: drop the ceiling lower in that room (section 5).
- Install the unit **level**. It must **never be covered** by insulation, with at least **15 mm** clear on the sides.
- Keep objects at least 1 m from the heat lamps. Hard-wired, with an isolating switch, by a licensed electrician.

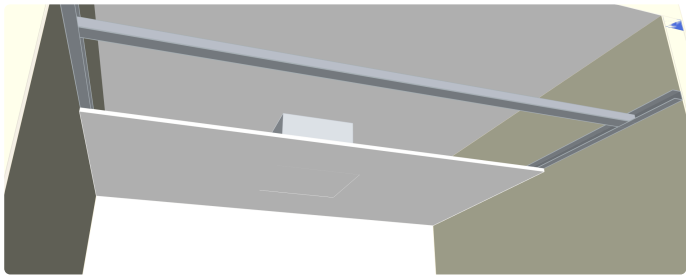
5 Rooms with a heat / fan / light unit: dropping the ceiling lower

A dropped ceiling held on the room's walls, like an insulated-panel dropped ceiling on a perimeter angle, but lined with lightweight plasterboard: wall track around the walls at the new height, ceiling channels spanning between, plasterboard underneath and a shadow line around the edge. Nothing is fixed to the roof panel. Check the span table of the ceiling system used against the room's width and the board thickness.

Wall to wall with a shadow-line edge, for rooms up to about 1.8 m across

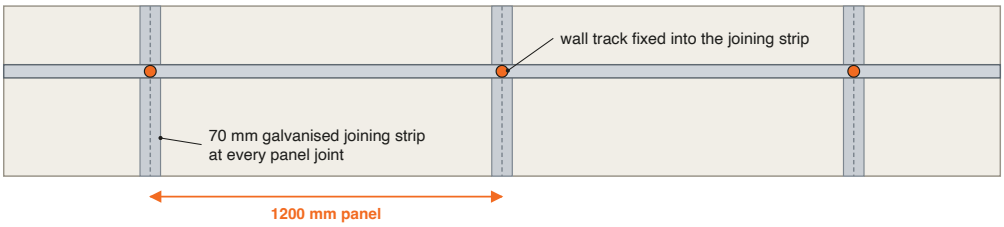


1. Wall track on two opposite walls
2. Ceiling channels spanning wall to wall
3. Plasterboard
4. Heat / fan / light unit
5. Space for the unit: IXL 210–245 mm
6. Shadow-line stopping bead at the board edge



Wall track, ceiling channels and the unit; nothing fixed to the roof panel.

Fixing the wall track (wall elevation)



- Fix wall track to two opposite walls at the new ceiling height. **Fix it into the 70 mm galvanised joining strip at every panel joint** (joints at 1200 mm centres): the strips give the structural fixing, not the 6 mm FC face and foam.
- Span 48 mm ceiling furring channels between the tracks: about 1500 mm with 16 mm plasterboard, up to 1750 mm with 10 or 13 mm.
- Up to about 1.8 m: 51 mm steel studs in matching track at 600 mm centres span just over 1800 mm with 10 mm board, about 1700 mm with 16 mm.
- Screw fix the ceiling channels to the wall track.
- Finish the plasterboard edge into a shadow-line stopping bead (10 × 10 mm recess, for boards up to 16 mm) for a shadow line around the ceiling.
- For a wider room, contact Quick Built Systems before planning a heat / fan / light unit.